

INVESTIGATION REPORT

Issue 323: zext.h wrongly rejected as illegal with the B extension (closed bug, replayed)

openhwgroup/cv32e40x

Document	RTL-TRIAGE-CV32E40X-323
Disposition	Bug confirmed, fix verified
Record date	2026-09-11
Classification	Internal working document. Not a verification sign-off.

1 **Status**

Issue	openhwgroup/cv32e40x#323
Issue type	Bug report from an integrator or user
Affects	cv32e40x
Configuration	B_EXT=ZBA_ZBB_ZBC_ZBS, X_EXT=0; pre-fix RTL e41bc6ee (2021-11-24) vs master
Disposition	Bug confirmed, fix verified
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	6 cited, 4 present, 2 not retained

Disposition meaning. The reported behaviour was reproduced, and the checks listed under Evidence pass against the corrected design.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 **Description**

Confirmed on the pre-fix RTL and verified fixed on master. Property: with the B extension enabled (B_EXT=ZBA_ZBB_ZBC_ZBS), a zext.h instruction (funct7=0000100, rs2=0, funct3=100, opcode=0110011) that retires in WB outside a debug entry must not raise an illegal-instruction exception; environment: no bus errors. On RTL e41bc6ee (2021-11-24) BMC finds an 8-cycle counterexample: zext.h (0x08004033) reaches WB at cycle 7 with exception_in_wb=1 and cause 2 (illegal instruction). On master the property is proven for all time (abc pdr, 6 s), holds in a 25-cycle BMC, and the cover shows zext.h retiring without exception at cycle 7. Comparison with the maintainers' fix, read afterwards (PR #328, merged 2021-11-25): the Zbb decoder gains the zext.h encoding (funct7 0000100 with rs2=0 routed to the pack/zext ALU op) plus an explicit illegal check when rs2 != 0. That is exactly the decode gap the counterexample shows. One iteration; the only extra work was a wrapper variant with the B_EXT parameter set on both RTL revisions.

3 **Conditions**

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

4 **Implications**

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Checker used

checker_323.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	counterexample	formal/prefix_bmc/job/engine_0/trace.vcd	not kept	pre-fix: zext.h in WB at cycle 7 with illegal-instruction exception (cex_prefix_zext.h.png)
E2	proof	formal/prefix_cover/sby.log	present	pre-fix: zext.h reaches WB (cover @8)
E3	proof	formal/master_prove/sby.log	present	master: abc pdr unbounded PASS (6 s)
E4	proof	formal/master_bmc/sby.log	present	master: BMC depth 25 PASS
E5	proof	formal/master_cover/sby.log	present	master: zext.h retires without exception at cycle 7
E6	diff	cores/cv32e40x_hist (git show ea79e0b9)	not kept	PR #328: b_decoder adds zext.h decode and rs2!=0 illegal check. Read after proving.

Paths are relative to runs/e40x-323. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.
- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- diff. Source or configuration difference.

Artefact(s) E1, E6 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

7 Limitations and scope

- The fix is verified against the checks listed under Evidence only. This is not a regression sign-off and does not cover the wider test suite.
- This is an agent investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

8 Provenance and measurement

Run directory	runs/e40x-323
Title source	Title and configuration read back from reports/issues/cv32e40x_issue_323.pdf, the previously published report for this issue.
Inputs shown	title, body
Human record	PR #328 (2021-11-25): zext.h added to the Zbb decoder (funct7 0000100, rs2=0), illegal when rs2 != 0
Agent confidence	0.92. This is the investigating agent's own estimate. It is not a measurement and not a reliability figure.
Iterations	1
Run by	Claude directly

Measured tool and step time

Step	Measured
read issue + pre-fix checkout	2.0 min (120 s)
B_EXT wrappers (generator tool) + checker	4.0 min (240 s)
pre-fix BMC (CEX) + cover	50.0 s
master BMC + cover + pdr	52.0 s
compare fix PR #328	60.0 s
report	60.0 s
Total	9.7 min (582 s)

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

Note. wall clock from worktree checkout to verdict; phases estimated from job timestamps

Generated by reporting/scripts/issue_document.py from runs/e40x-323/verdict.json. The generator typesets recorded data and checks that cited files exist. It does not run, rerun or validate any check, and it cannot supply missing evidence.